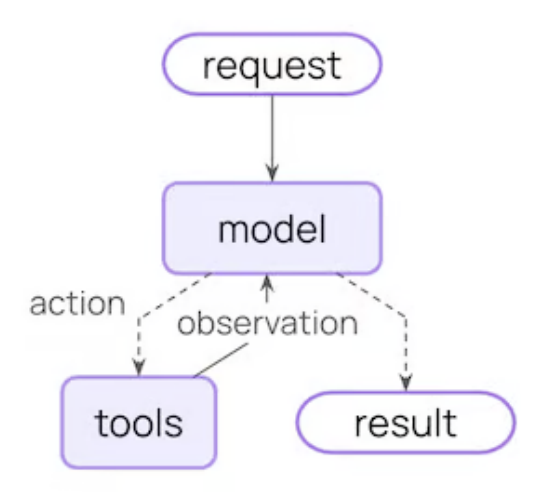
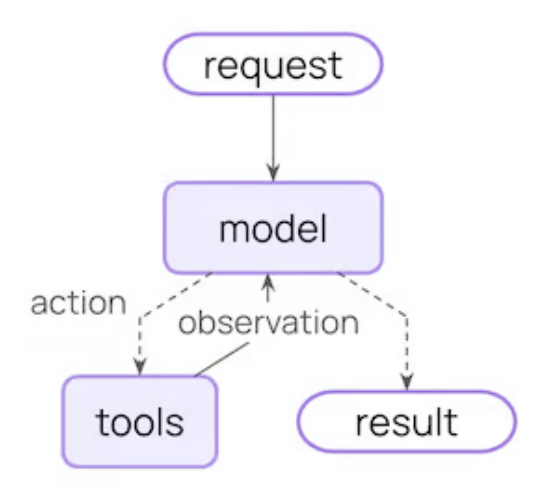

request
model
action
observation
tools
result

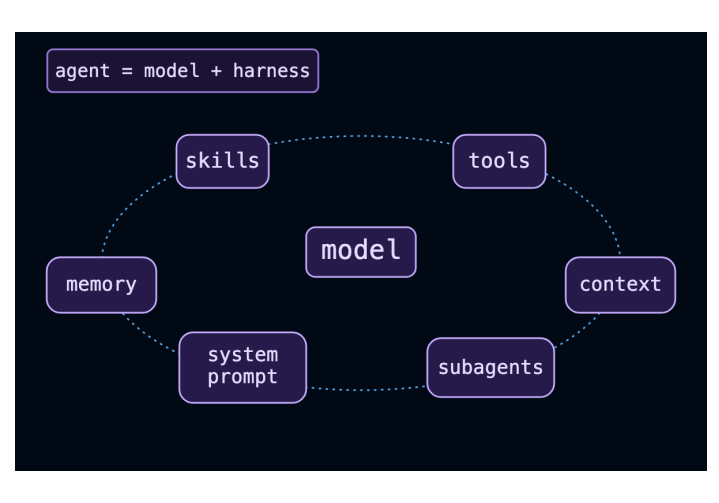
Cinebot

TripMate

Middleware

Middleware provides us a way to more tightly control what happens inside the agent.


request
model
action
observation
tools
result


agent = model + harness
skills
tools
model
memory
context
system prompt
subagents

Talk in a rude manner →

give some personal info

→ **Abuse or talk badly as well.**

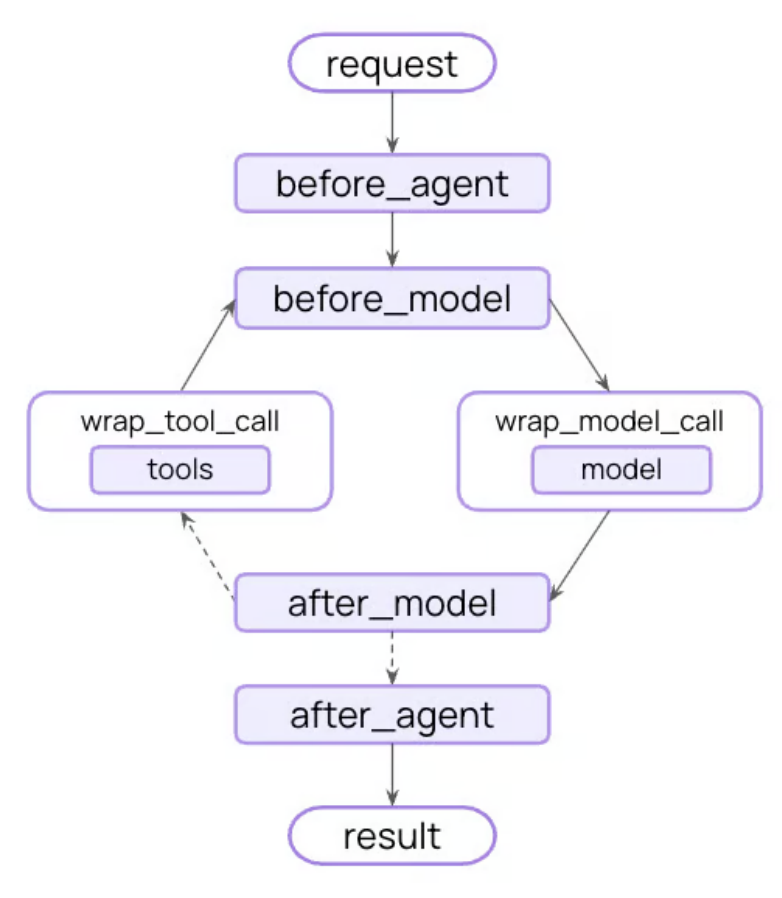
Flagging the same.

Tracking agent behavior with logging, analytics, and debugging.

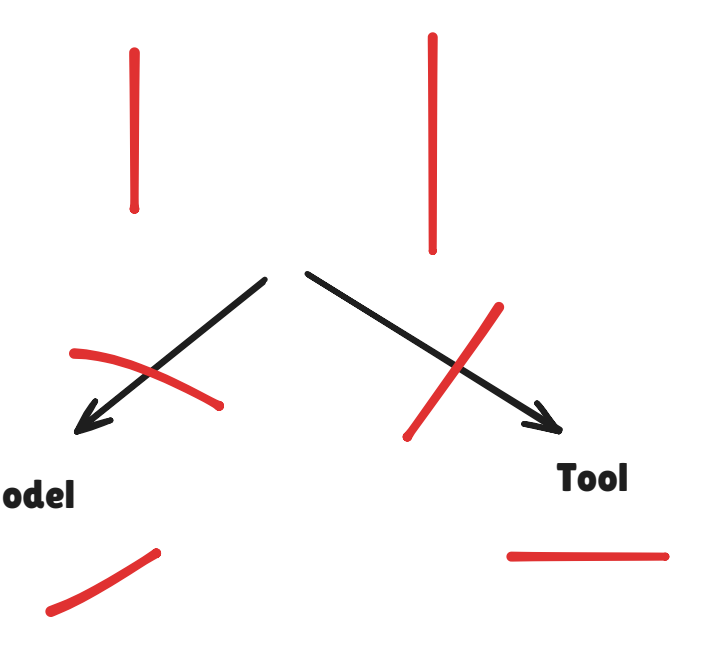
Transforming prompts, tool selection, and output formatting.

Adding retries, fallbacks, and early termination logic.

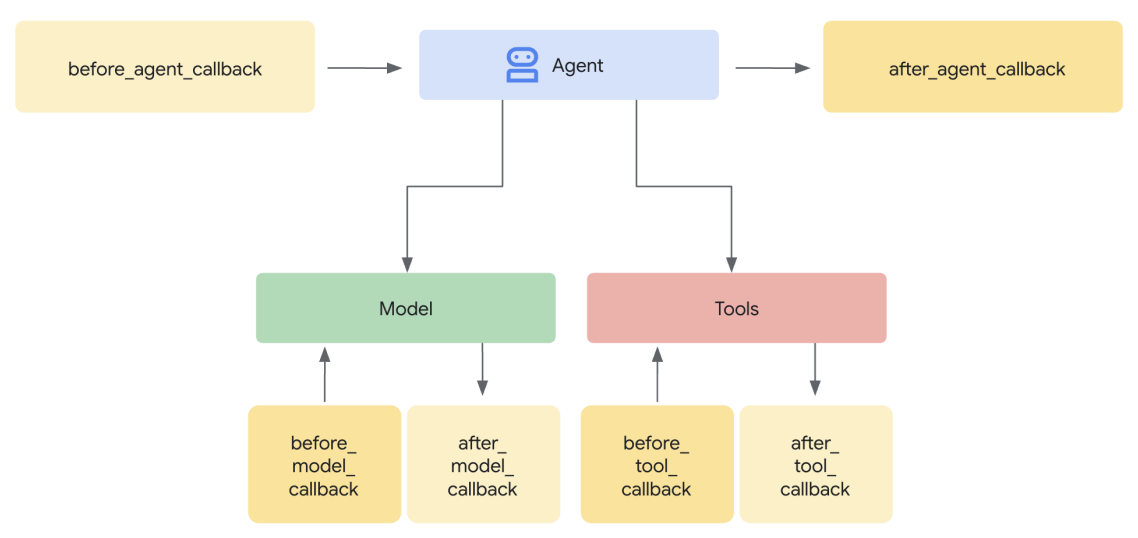
Applying rate limits, guardrails, and PII detection.

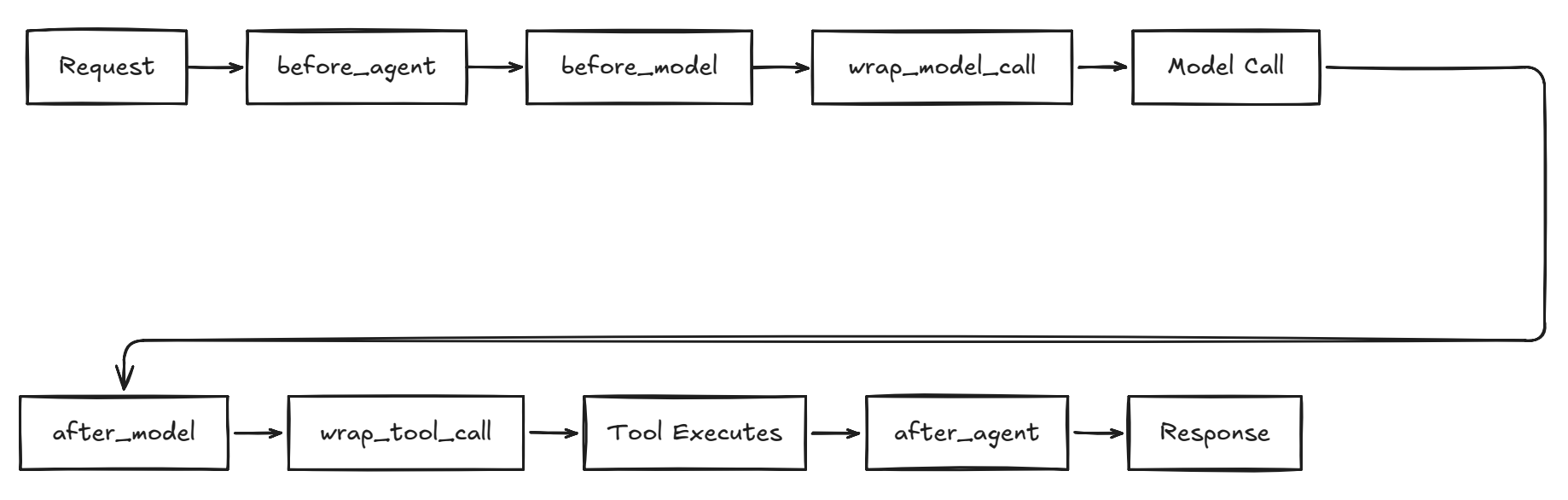

request
before_agent
before_model
wrap_tool_call
tools
wrap_model_call
model
after_model
after_agent
result

Human in the Loop
Guardrails
Summarization
Dynamic Tool Loading.



agent
model
tool


before_agent_callback
Agent
after_agent_callback
Model
Tools
before_model_callback
after_model_callback
before_tool_callback
after_tool_callback

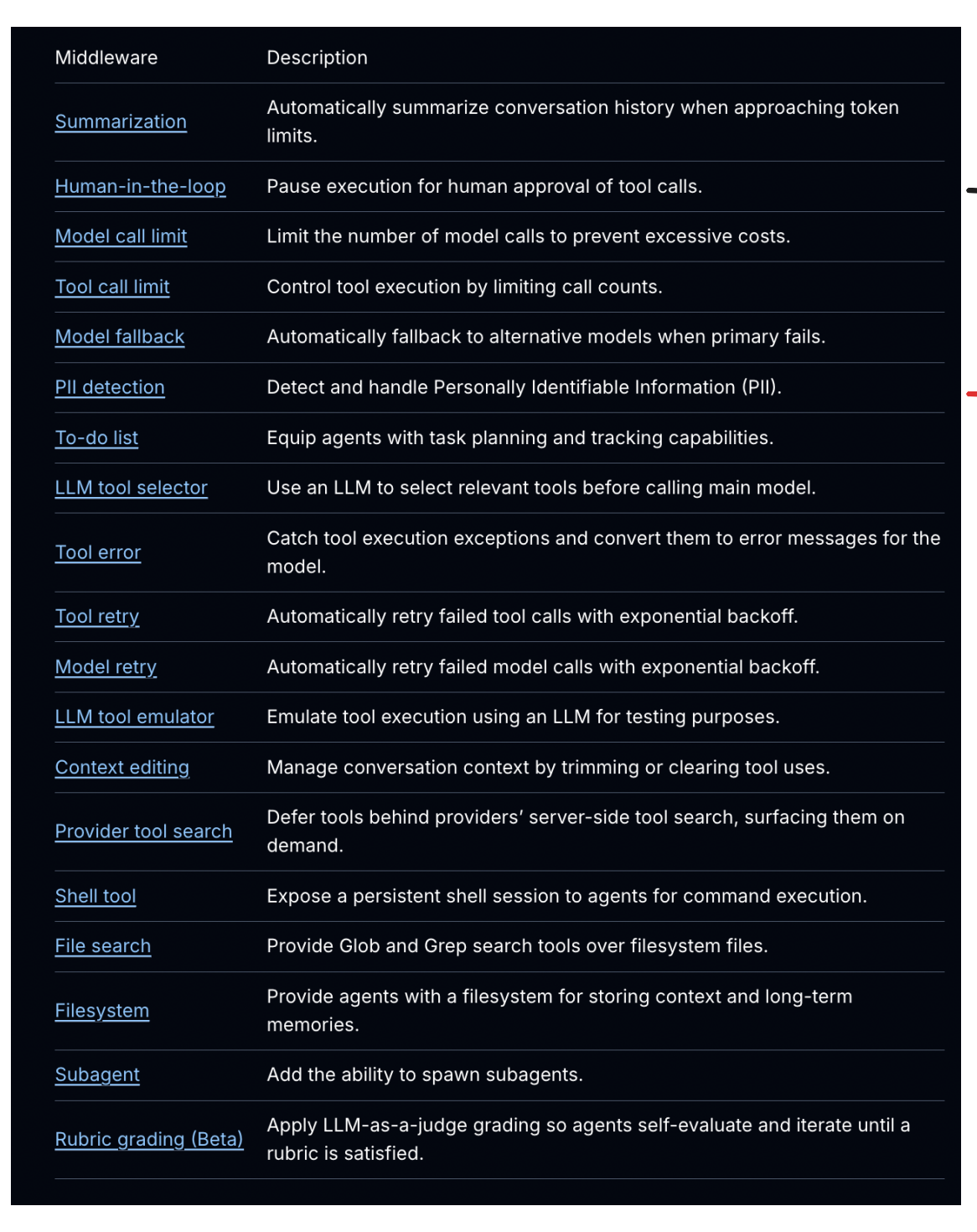

Request
before_agent
before_model
wrap_model_call
Model Call
after_model
wrap_tool_call
Tool Executes
after_agent
Response

Document

Tripmate

Middleware → **Prebuilt** →

Middleware → **Custom**

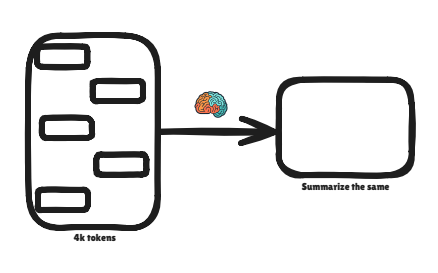


Middleware	Description
Summarization	Automatically summarize conversation history when approaching token limits.
Human-in-the-loop	Pause execution for human approval of tool calls.
Model call limit	Limit the number of model calls to prevent excessive costs.
Tool call limit	Control tool execution by limiting call counts.
Model fallback	Automatically fallback to alternative models when primary fails.
PII detection	Detect and handle Personally Identifiable Information (PII).
To-do list	Equip agents with task planning and tracking capabilities.
LLM tool selector	Use an LLM to select relevant tools before calling main model.
Tool error	Catch tool execution exceptions and convert them to error messages for the model.
Tool retry	Automatically retry failed tool calls with exponential backoff.
Model retry	Automatically retry failed model calls with exponential backoff.
LLM tool emulator	Emulate tool execution using an LLM for testing purposes.
Context editing	Manage conversation context by trimming or clearing tool uses.
Provider tool search	Defer tools behind providers' server-side tool search, surfacing them on demand.
Shell tool	Expose a persistent shell session to agents for command execution.
File search	Provide Glob and Grep search tools over filesystem files.
Filesystem	Provide agents with a filesystem for storing context and long-term memories.
Subagent	Add the ability to spawn subagents.
Rubric grading (Beta)	Apply LLM-as-a-judge grading so agents self-evaluate and iterate until a rubric is satisfied.

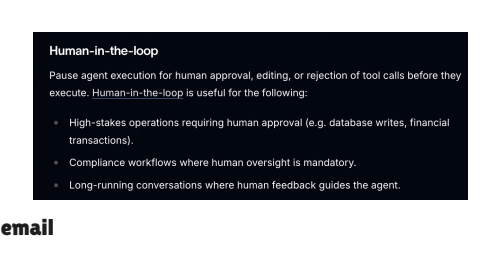
→ **HITL**

→ **PII** → **Email Mobile Number**

PII → **Adhaar PAN**


4k tokens
Summarize the same

100k ----> 10k tokens



Human-in-the-loop

Pause agent execution for human approval, editing, or rejection of tool calls before they execute. Human-in-the-loop is useful for the following:

- High-stakes operations requiring human approval (e.g. database writes, financial transactions).
- Compliance workflows where human oversight is mandatory.
- Long-running conversations where human feedback guides the agent.

Intern

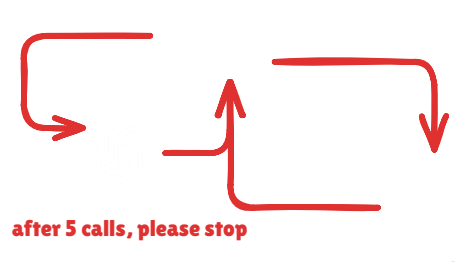
send_email **read_email**

```
middleware=[
    HumanInTheLoopMiddleware(
        interrupt_on={
            "your_send_email_tool": {
                "allowed_decisions": ["approve", "edit", "reject"],
            },
            "your_read_email_tool": False,
        }
    ),
],
```

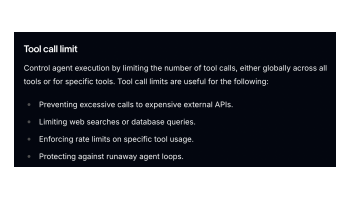
HITL middleware allows us to have a human oversight when an agent is calling a tool.

Request ---> Brain ----> Tool

HITL



after 5 calls, please stop



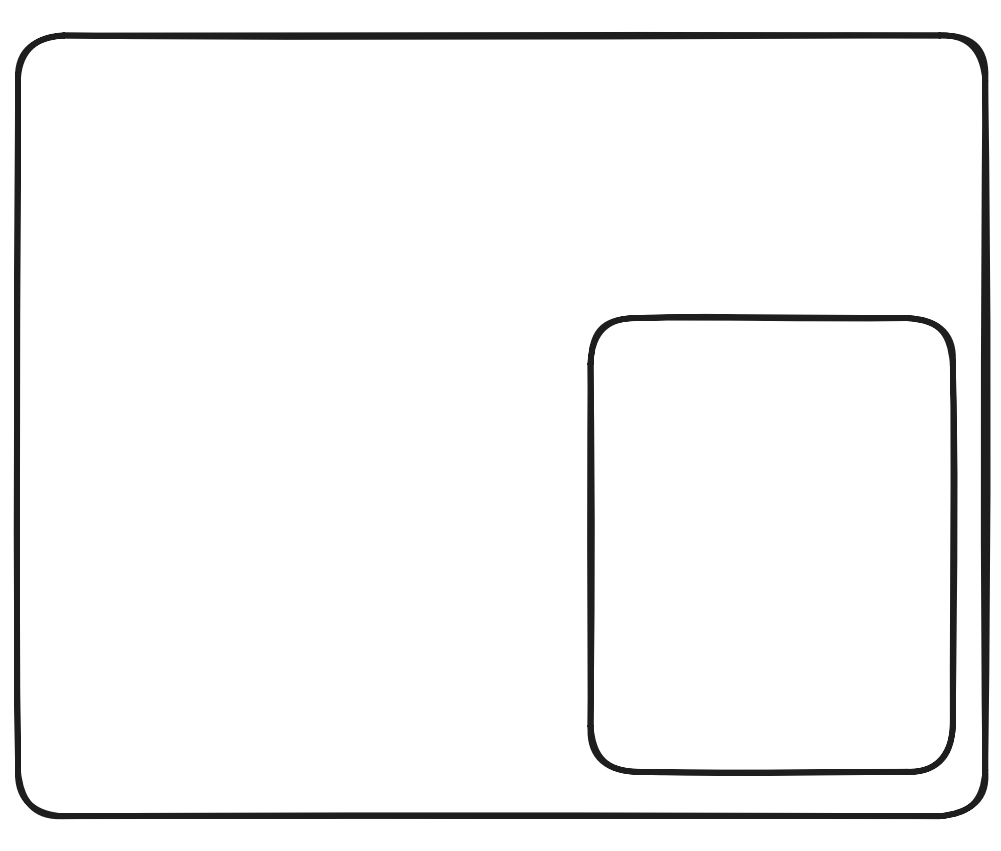
Tool call limit

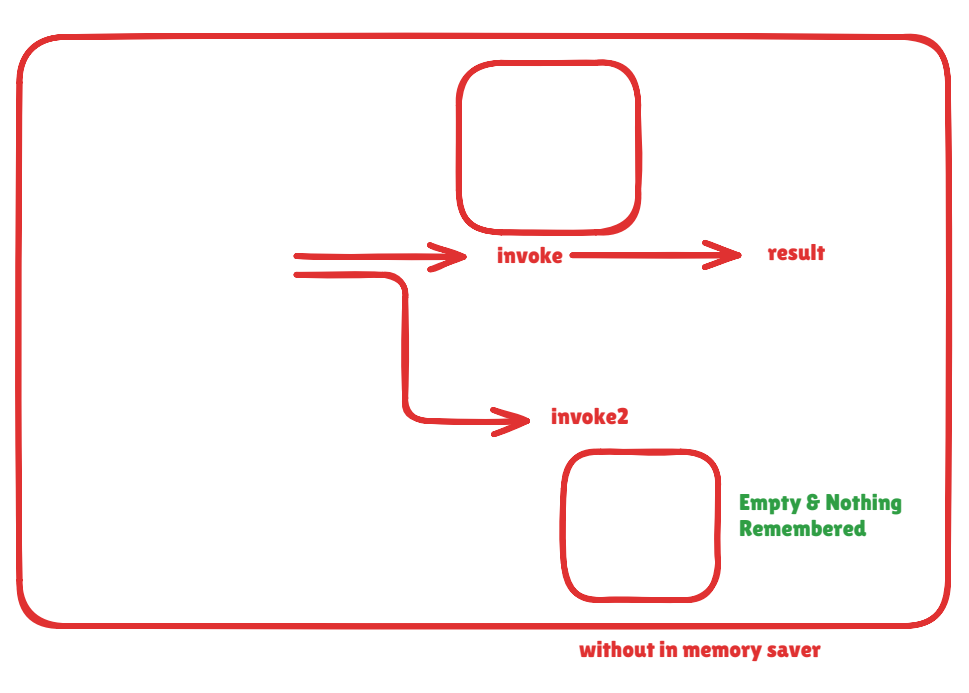
Control agent execution by limiting the number of tool calls, either globally across all tools or for specific tools. Tool call limits are useful for the following:

- Preventing excessive calls to expensive external APIs.
- Limiting web searches or database queries.
- Enforcing rate limits on specific tool usage.
- Protecting against runaway agent loops.

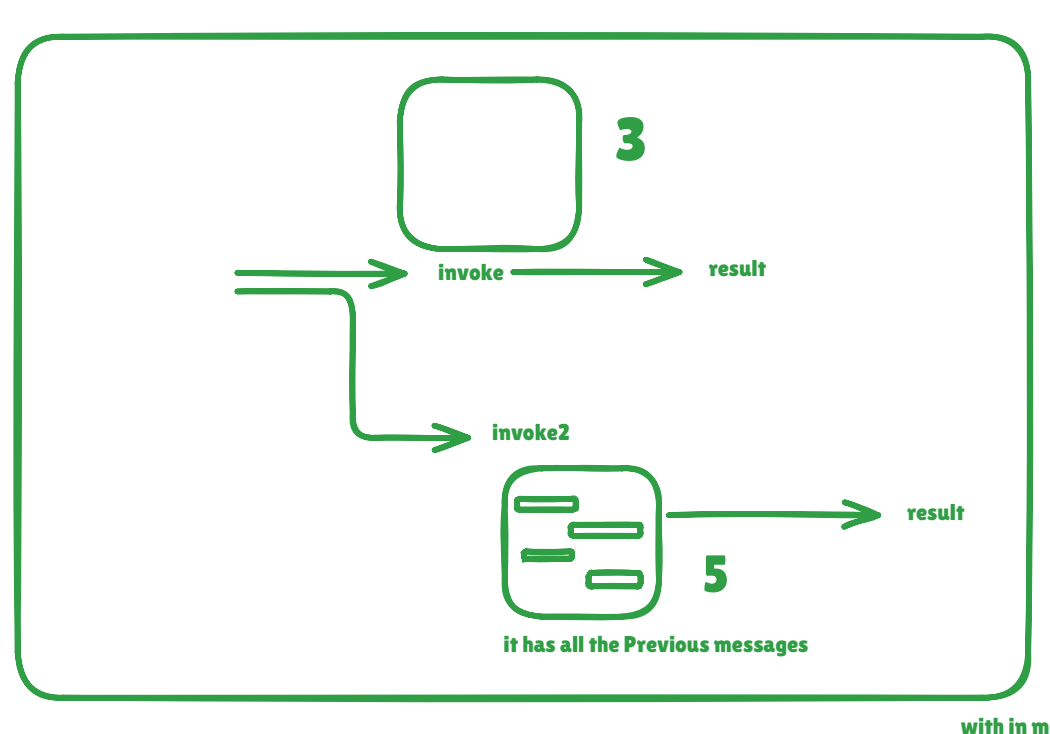
Tool Call limit

Middleware
Hooks
Callbacks




invoke
result
invoke2
Empty & Nothing Remembered

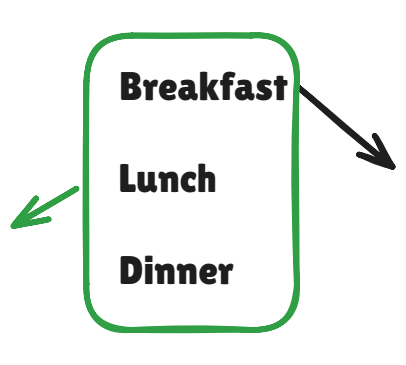
without in memory saver


3
invoke
result
invoke2
result
5
it has all the Previous messages

with in memory saver

Run limit – on a single invoke

Thread Limit – Across all runs.


Breakfast
Lunch
Dinner

run limit

Chappatis allowed per meal

thread limit

Chappatis allowed total across all meals

For a specific Tools

CancelBooking

(run _limit) **1**

2 (Thread or Conversation Limit)

For All Tools

8

B100 → **run_limit = 1 thread_limit = 1**

B101 → **run_limit = 1 thread_limit = 2**

Got Refreshed

B102 → **run_limit =**

Guardrails is a way to control the AI agent behaviour

PII – Personal Identifiable Information

Check my calendar ?

Drive

Calendar

Calendar

Gmail

Slack

System Prompt

Tool Description

Can you cancel my booking 12345.

HumanMessage
AIMessage
ToolMessage
AIMessage

Initial_Delay =

→ → **4**

```
    ),
    ToolMessage(
        content="Tool 'flaky_showtime_check' failed after 4 attempts with ConnectionError: Simulated network failure -- exactly what a real external call risks.. Please try again.",
        name='flaky_showtime_check',
        id='72161a9c-cdf5-4109-a46d-66cdff0fedfc',
        tool_call_id='call_MYYPxrSFyudgrvWcSRP8sOul',
        status='error'
    )
```

→ → **4 (1 default and 3 retries)**

AI –> I am unable to connect.